



RAMCO INSTITUTE OF TECHNOLOGY

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Department of Information Technology Academic Year 2024 - 2025 (Even Semester)

Degree, Semester & Branch: IV Semester B.Tech. IT

Course Code & Title: CS3491 & Artificial Intelligence and Machine Learning

Name of the Faculty member (s): Mrs. G.Mareeswari, AP/IT

Innovative Practice Description

- **Unit / Topic:** Unit IV / Boosting
- **Course Outcome:** CO4
- **Topic Learning Outcome:** TLO16
- **Activity Chosen:** One Minute Paper

- **Justification:**

Boosting is a popular ensemble learning technique that improves the performance of machine learning models by combining multiple weak learners into a strong one. It focuses on correcting errors made by previous models, making it highly effective for classification and regression tasks. Boosting algorithms like AdaBoost, Gradient Boosting, and XGBoost are widely used in real-world applications and machine learning competitions. This topic is relevant and impactful for anyone interested in enhancing their model's accuracy and understanding key machine learning techniques. This activity makes the students to quickly grasps the importance of boosting techniques in improving the performance of machine learning models

- **Time Allotted for the Activity:** 10 Minutes

- **Details of the Implementation:**

At the end of the class, students were asked to write about the Boosting topic discussed. The instructor explained key concepts such as combining weak learners into a strong learner and algorithms like AdaBoost, Gradient Boosting, and XGBoost. Students shared their understanding and areas needing clarification, focusing on how boosting improves model accuracy. The feedback highlighted their comprehension of boosting and its applications in machine learning. This activity served as a quick check to assess their understanding and engagement with the topic. The points on Boosting topic shared by the students through this activity as shown in the figure 1a and figure 1b.

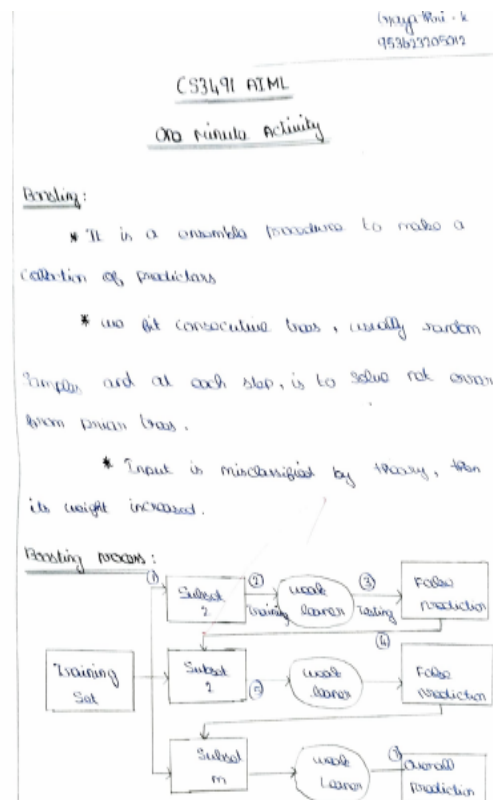
• CO – PO / PSO mapping:

CO	PO1	PO2	PO3	PO9	PSO1
CO4	3	1	1	1	1
(1 – Low 2 – Moderate 3 – High)					

• PO / PSO mapped:

Innovative practice	PO1	PO2	PO3	PO9	PSO1
CO4	3	1	1	1	1
Justification for correlation	Needs engineering and mathematical knowledge to understand Boosting concept	Various boosting techniques used in analysis of solving complex engineering problems	The students can able to find solution for the complex problems using the concept of Boosting.	The students individually involved in this activity	The students can apply the concept of Boosting in solving various problems of data science and algorithms.

• Images / Screenshot of the practice:



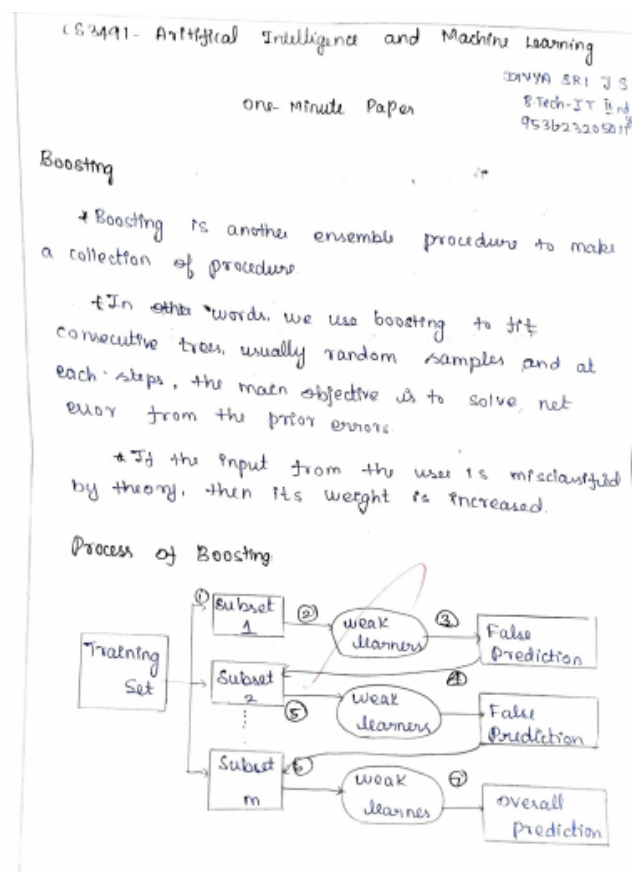


Figure 1. a Sample One-Minute Paper of Divya Sri.J.S and Gayathri. K

CS3491 AIML

ONE MINUTE ACTIVITY

Boosting:

- * It is a sequential procedure to make a collection of predictors.
- * We fit consecutive trees, usually random samples and at each step, is to solve not learn from prior trees
- * Input is misclassified by theory, then its weight increased.

Boosting Process:

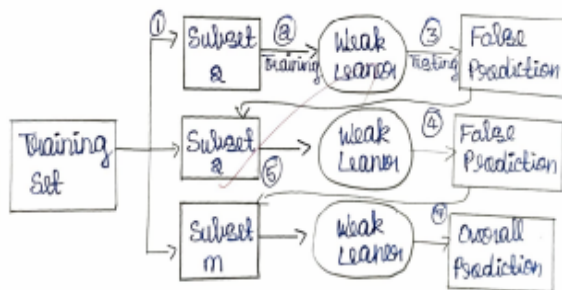


Figure 1.b Sample One-Minute Paper of Vishal Kumar. S and Arjun. G

Reflective Critique:

❖ **Feedback of practice from students and other stakeholders:**

- The students mentioned that the activity encouraged them to actively engage with the topic and summarize it in their own words.
- The students felt more confident in recalling the key concepts discussed during the session after completing this activity.

❖ **Benefit of the practice:**

- It helps students to explain the topic in their own words, which makes learning easier.
- Teachers can quickly understand what students have understood and what they haven't.
- Students can find out what they know well and what they need to concentrate more.

❖ ***Challenges faced in implementation:***

- Some students understand the concept but struggle to write it down clearly.
- Due to time constraints some students not drawn the architecture of boosting concept.

References:

- ❖ https://www.ritrjpm.ac.in/images/B.Tech-IT/2024-2025/GM_CD3291_One_Minute_Paper.pdf
- ❖ <https://www.rochester.edu/college/cetl/faculty/one-minute-paper.html>
- ❖ <https://www.unl.edu/gradstudies/current/teaching/minute>
- ❖ https://www.ritrjpm.ac.in/images/computer-science/2022-2023/1_GM_CS335_3_Functions_OMP.pdf
- ❖ <https://oncourseworkshop.com/self-awareness/one-minute-paper/>

Signature of Faculty Member

HOD